Question ID 1b2b20b9

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	

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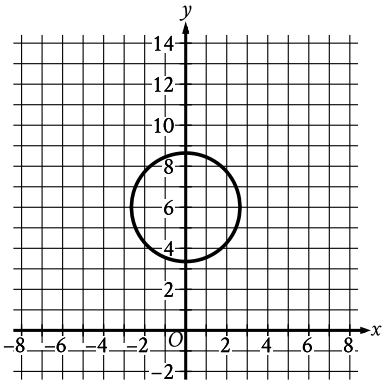

Circle A shown is defined by the equation $x^2 + (y-6)^2 = 7$. Circle B (not shown) has the same radius but is translated 96 units to the right. If the equation of circle B is $(x-h)^2 + (y-k)^2 = a$, where h, k, and a are constants, what is the value of $4a$?

ID: 1b2b20b9 Answer

Correct Answer:

28

Rationale

The correct answer is 28. The equation of a circle in the *xy*-plane can be written as $(x-t)^2 + (y-s)^2 = r^2$, where the center of the circle is (t, s) and the radius of the circle is r. It's given that circle A is defined by the equation $x^2 + (y-6)^2 = 7$, which can be written as $(x-0)^2 + (y-6)^2 = \left(\sqrt{7}\right)^2$. It follows that $r = \sqrt{7}$ and the radius of circle A is $\sqrt{7}$. It's also given that circle B has the same radius as circle A. If the equation of circle B is $(x-h)^2 + (y-k)^2 = a$, then $a = r^2$. Substituting $\sqrt{7}$ for r in this equation yields $a = \left(\sqrt{7}\right)^2$, or $a = 7$. It follows that the value of $4a$ is $4(7)$, or 28.

Question Difficulty:

Hard